

SABIN PYAKUREL

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PROFILE

Full-stack engineer specializing in deploying AI features into production systems using Python and modern JavaScript frameworks. Experienced in building intelligent systems using NLP embeddings, LLM-powered assessment platforms, and workflow automation. Comfortable shipping end-to-end features, working with teams to produce test-safe systems. Extremely adaptable to new technological changes.

EDUCATION

Bachelor of Computer Engineering | Top 5% in academic cohort
Tribhuvan University

CERTIFICATIONS & BADGES

AWS Knowledge: Serverless | *Amazon Web Services Training Badge*

PROJECTS

AI-Powered Resume Intelligence & RAG System | *Django, Python, NLP, Embeddings*

- Built LLM-powered resume intelligence system using Word2Vec embeddings for semantic search, achieving 80% matching accuracy across 8 job categories(backend, ML engineer) , a 3× improvement over keyword-only TF-IDF baseline.
- Implemented GPT-2 perplexity scoring to automatically flag AI-generated or keyword-stuffed resumes, reducing manual screening effort by an estimated 60% for borderline submissions over 100 samples.
- Designed RAC-style retrieval pipeline using cosine similarity on dense embeddings, enabling sub-second candidate ranking over a corpus of 500+ resumes.

Proctored Technical Assessment Platform | *Django, Docker, JavaScript*

- Programmed multi-stage assessment platform (MCQ, coding, subjective, aptitude) with dynamic question delivery, supporting 4 distinct evaluation pipelines within a single unified system.
- Integrated Flan-T5-large LLM for automated reference answer generation using few-shot prompting with constrained output, full automating answer evaluation time for recruiters.
- Implemented identity verification using dlib facial recognition (68-point landmarks, ResNet embeddings), maintaining under 5% false-acceptance rate across test sessions.
- Used Docker-based sandboxing to securely execute and evaluate code submissions by creating test-cases in isolated environments, eliminating host-system exposure across all candidate runs.

Hiring Automation & Notification Pipeline | *Django, Celery, Redis*

- Engineered asynchronous workflow system for automated candidate notifications and bulk email processing, handling 200+ emails per hiring cycle without manual intervention.
- Offloaded background jobs to Celery and Redis, reducing API response times from ~3s to under 300ms under concurrent load.
- Designed token-based email workflow that removed manual follow-up steps for 100% of standard hiring communication touchpoints.

ADDITIONAL PROJECTS

Unity Game Projects | *Unity, C#*

- Built interactive 2D game systems with UI flow, gameplay loops, and state management.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C#, SQL

Backend / Systems: Django, Django REST Framework, Celery, Redis, Docker, REST API design

Frontend / Full-stack: React, HTML, CSS, JavaScript

GenAI / LLM: Word2Vec, GPT-2, Flan-T5, Transformers, Embeddings, RAC, Prompt Engineering

ML / NLP: TF-IDF, SVM, NLTK, scikit-learn, NumPy, pandas, dlib

Tools & Cloud: Git, GitHub, FFmpeg, AWS (Serverless)